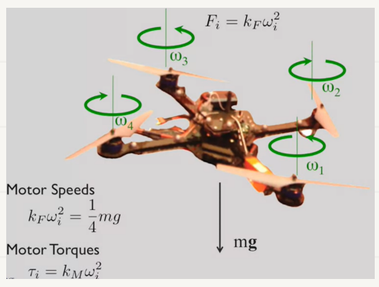
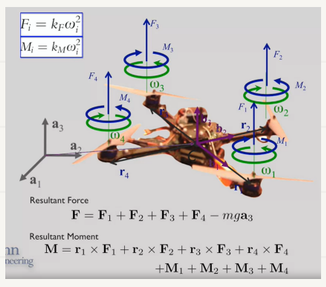
1.2 Energetics and System Design

Basic Mechanics

- Basics - Control - Design considerations - Agility - Component selection - Effect of size





Dynamics and 1-D Linear Control

Control of height

$\sum_{i=1}^{4} k_F \omega_i^2 + mg = ma$

$a = \frac{d^2x}{dt^2} = \ddot{x}$

Input $u = \frac{1}{m}\left[\sum_{i=1}^{4} k_F \omega_i^2 + mg\right] = \ddot{x}$ (Second order dynamic system)

Control of a linear second-order system

Problem: State, input: $x, u \in \mathbb{R}$

Plant model: $\ddot{x} = u$

Find a control input function $u(t)$ so that $x(t)$ follows the desired trajectory $x^{des}(t)$

General approach: Define error: $e(t) = x^{des}(t) - x(t)$ $\quad e(t) \to 0$ (渐近稳定)

Strategy: Find $u(t)$ such that $\ddot{e} + k_v \dot{e} + k_p e = 0$ $\quad k_p, k_e > 0$

PD control: $u(t) = \underbrace{\ddot{x}^{des}(t)}_{\text{Feedforward}} + \underbrace{k_v \dot{e}(t)}_{D} + \underbrace{k_p e(t)}_{P}$

PID control: $u(t) = \ddot{x}^{des}(t) + k_v \dot{e}(t) + k_p e(t) + \underbrace{k_i \int_0^t e(\tau)\, d\tau}_{I}$ 补偿未知扰动

Design Considerations

$u = \frac{1}{m}\left[\sum_{i=1}^{4} k_F \omega_i^2 + mg\right] = \frac{1}{m}[T + mg]$

推重比 $T_{max} \to u_{max}$

PD control: $u(t) = \min(\ddot{x}^{des}(t) + k_v \dot{e}(t) + k_p e(t), u_{max})$

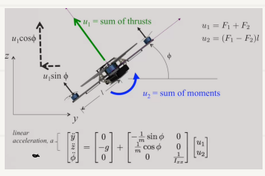
功耗 电池选择

Batterys ~ 33% mass $\quad$ Motors ~ 25 mass

Agility and Maneuverability 创建敏捷机器人

减少制动距离

快速制动，快速转弯



Two key ideas

- Accelerate quickly maxize a_{max} 线 maximize $\frac{u_{1,max}}{m}$
- Roll / pitch quickly maxize α_{max} 角 maximize $\frac{u_{2,max}}{I_{xx}}$

Component Selection.

	FRAME + PIXHAWK + PROPULSION	BATTERY	PAYLOAD	TOTAL WEIGHT	MAX THRUST	THRUST/ WEIGHT	Propeller
3DR X8+	1855	817	600	3272	10560	3.227	11" x 4.7"
DJI F550 + E600	1494	721	600	2815	9600	3.410	12" x 4.2"
DJI F450 + E310	826	400	600	1826	3200	1.752	9.4" x 5"
DJI F450 + E600	970	721	600	2291	6400	2.794	12" x 4.2"
DJI F550 + E310	1278	400	600	2278	4800	2.107	9.4" x 5"
DJI F550 + E310 @ 4 cell	1278	600	600	2478	5316	2.145	9.4" x 5"
DJI F550 + E305 @ 4 cell	1124	600	600	2334	5100	2.185	9.4" x 5"

Pixhawk $\quad$ Intel NUC 记

Laser scanner $\quad$ Cameras

Effects of Size

$m \sim l^3 \quad I \sim l^5$ mass, inertia $\qquad$ Since $r \sim l$

$F \sim \pi r^2 \times (wr)^2 = v^2 r^2$ thrust $\qquad F \sim l^2 v^2 \quad a \sim \frac{F}{m} \quad a \sim \frac{v^2}{l}$

$M \sim Fl$ moment $\qquad M \sim l^3 v^2 \quad \alpha \sim \frac{M}{I} \quad \alpha \sim \frac{v^2}{l^2}$

Agility with Scaling

Froude scaling: $v \sim \sqrt{l}$ $\qquad$ Mach scaling: $v \sim 1$

$F \sim l^3 \qquad F \sim l^2$

$a \sim 1 \quad \alpha \sim \frac{1}{l} \qquad a \sim \frac{1}{l} \quad \alpha \sim \frac{1}{l^2}$

Supplementary Material: Introduction

- Hybrid Dynamical Systems 混杂动态系统.
- High-Order Under Actuated Systems 高阶欠驱动系统
- Multi-Robot systems 多机器人系统.

Supplementary Material: Dynamical Systems

Supplementary Material: Rates of Convergence